$$= \text{Antilog } \frac{(6.1353)}{3}$$

$$= \text{Antilog } (2.0451)$$

$$= 110.9$$

## (b) G.M. For Discrete grouped data

If $x_1, x_2, \ldots\ldots x_n$ are discrete values of the variate x with corresponding frequencies $f_1, f_2, \ldots f_n$. Then geometric mean is defined as

$$\text{G. M. } = \text{Antilog } \frac{\sum_{i=l}^{n} f_i \log x_i}{N} \text{ with usual notations}$$

### Example 5.9

Find the G.M for the following data, which gives the defective screws obtained in a factory.

| Diameter (cm) | 5 | 15 | 25 | 35 |
|---|---|---|---|---|
| Number of defective screws | 5 | 8 | 3 | 4 |

***Solution:***

| $x_i$ | $f_i$ | $\log x_i$ | $f_i \log x_i$ |
|---|---|---|---|
| 5 | 5 | 0.6990 | 3.4950 |
| 15 | 8 | 1.1761 | 9.4088 |
| 25 | 3 | 1.3979 | 4.1937 |
| 35 | 4 | 1.5441 | 6.1764 |
| | N=20 | | 23.2739 |

$$\text{G.M } = \text{Antilog}$$

$$= \text{Antilog } \frac{\sum_{i=l}^{n} f_i \log x_i}{N}$$

$$= \text{Antilog } \frac{23.2739}{20}$$

$$= \text{Antilog } 1.1637$$

$$\text{G.M } = 14.58$$



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